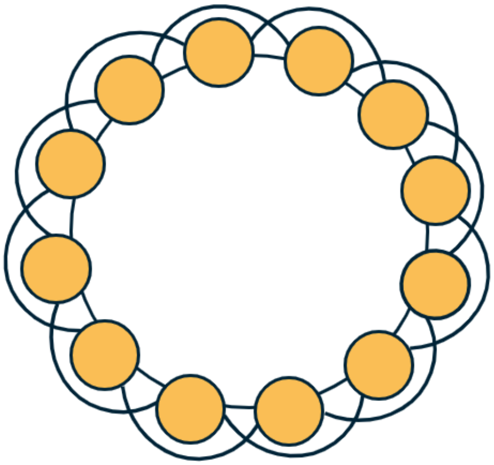
L5: Clustering in Regular Networks



On the other hand, regular networks typically have a locally clustered topology. The exact value of the clustering coefficient depends on the specific network type but in general, it is fair to say that ***"regular networks have strong clustering"***. Further, the clustering coefficient of regular networks is typically independent of their size.

To see that, let's consider the common regular network topology shown at the visualization. The *n* nodes are placed in a circle, and every node is connected to an even number *c* of the nearest neighbors *(c/2 at the left and c/2 at the right).* If *c=2*, this topology is simply a ring network with zero clustering *(no triangles)*. For a higher *(even)* value of *c* however, the transitivity coefficient is:

$T = \frac{3(c-2)}{4(c-1)}$

Note that this does not depend on the network size. Additionally, as *c* increases the transitivity approaches ¾.

Food For Thought

Prove the previous formula for the transitivity coefficient.